# Assignment 1: Fibonacci in LC-3 Assembly

## Overview

In this assignment, you will write an LC-3 Assembly program to calculate the **20th number of the Fibonacci sequence**. If possible, extend your program to calculate the **nth number** of the sequence. Use the **5-step approach** to design and implement your code, and run it using LC3Tools.

## Requirements

- Write an LC-3 Assembly program to calculate the 20th Fibonacci number.
- Enhance the program to handle input and calculate the nth Fibonacci number (optional).
- Follow the 5-step approach to structure your program.
- Run and test your program on LC3Tools.
- Submit both your **code file** and a **screenshot of LC3Tools output**.

## Deliverables

- LC-3 Assembly source code file.
- Screenshot of program execution in LC3Tools.
- Optional enhancements for calculating the nth Fibonacci number.

## Evaluation Criteria (100 points)

| Category | Description | Points |
|---|---|---|
| Correctness | Program correctly calculates the 20th Fibonacci number. | 30 |
| Extension | Ability to calculate nth Fibonacci number (optional). | 20 |
| Code Structure | Proper use of 5-step approach in coding. | 20 |
| Testing & Output | Successful run and screenshot in LC3Tools. | 20 |
| Clarity | Code readability, comments, and submission format. | 10 |

## Submission

- Assignment releases tonight.
- Submit your work before **23:50 on Sunday (10/05)**.
- Collaborate with classmates if needed, and attend **office hours** for help.